

Core-IX: Web Development with PHP

Course Objectives:

- To understand the essentials of Server-side programming
- To understand web development using PHP

Learning Outcomes:

On successful completion of this course, Students will be able to:

- Learn the basics of JSON, XML and AJAX
- Learn the programming concepts of PHP
- Learn the server-side programming using PHP
- Learn the mechanisms of connecting Database using PHP & use AJAX with PHP

Unit I:

Introduction to Server Side Technologies, Web Servers, Understanding the concepts of JSON, AJAX: Introduction, Creating Internet Applications using AJAX. XML: Introduction, Features, Fundamentals, Document Type Definition, XML Schema.

Unit II:

PHP: Features, Programming fundamentals: Print/echo statement, Data Types, Variables, Constants, Strings, Arrays, Operators. Control Structures: Conditional, Looping & Jump Statements. Functions: String, Date-Time, Mathematical and User-defined functions. Embedding PHP in HTML, Reading Form data of a Web Page.

Unit III:

Introduction to PHP with Database: Connecting to Database, Selecting a Database, Adding Table and Altering a Table in a Database. Inserting Data, Modifying Data in a Table, Retrieving Data from a table and displaying in HTML.

Unit IV:

State Management in PHP: Introduction, Cookies, Session. Authentication in PHP: Creating a User, Adding authorized users, Displaying the User. Using AJAX: AJAX with PHP, AJAX with Database.

Text Book:

- ✓ *Web Technologies (Black Book), DreamTech Press*

Reference Books:

- ✓ *Web Enabled Commercial Application Development Using HTML, JavaScript, DHTML and PHP 4th Edition by Ivan Bayross.*
- ✓ *PHP and MySQL Web Development by Luke Welling and Laura Thomson. Addison - Wesley.*

Core-IX Lab: Web Development with PHP

1. Write PHP program (s) for the following.
 - a. Find greatest among three numbers entered by the user
 - b. Print the sum of numbers from M to N where their values are entered by the user.
 - c. Find the factorial of a number entered by the user.
2. Write a PHP program that asks the name and date of birth of the user.
 - a. Find the number of letters, words in the name
 - b. Display the Name in reverse order
 - c. Print the current date and time and age of the user.
3. Design a web page to create a form that collects the name, gender and mail of a person. Write a PHP program that collects the data entered by the user in the form and displays them in a new page.
4. Write a PHP program that creates a Table in a database. The number of columns of the table are determined by the fields in the form (created in question no. 3).
5. Write a PHP program to
 - a. insert new records
 - b. update a record
 - c. delete a record based on a value of a field in the table.
6. Write a PHP program that asks the user to enter a name and display the details of the user retrieved from the database in the same page. [show the error message if no matching name is found in the database].
7. Write a PHP program to create a cookie and store your name and then read the cookie.
8. Write a PHP program that allows only authenticated users to retrieve the details of a table. [Use username and password of the user to validate the authenticity].
9. Write a PHP application to make use of AJAX.